

Project Development Roadmap/Plan

Integrated Framework for Groundwater Recharge, Contaminant Transport and Potable Water Assessment

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Overall Project Plan

WP0

Before the main packages, we can look into conceptual & mathematical Foundations, analysis and verification of fracture porous media modeling or models

WP0 Objectives

Item	Description
Goal	Develop the conceptual framework of multi-continuum groundwater systems.
Scientific Question	How should unresolved heterogeneity be represented?
Tool	Mathematical model

WP0 Development Plan

Stage	Task	Description	Expected Deliverable
WP0.1	Literature Review	Review fractured porous media, dual-permeability, multi-continuum methods	Literature review report
WP0.2	Concept Development	Define continuum concept	Conceptual framework
WP0.3	Exchange Mechanisms	Develop continuum interaction equations	Exchange model
WP0.4	Governing Equations	Develop generalized Richards equation	Mathematical formulation
WP0.5	Closure Relations	Hydraulic conductivity, retention, exchange laws	Closure equations
WP0.6	Numerical Formulation	Discretization strategy	Numerical framework
WP0.7	Verification	Compare with analytical solutions	Verification report

WP0.5

Synthetic Benchmarks

Stage	Task	Deliverable
1	Create homogeneous benchmark	Test Case A
2	Layered soil	Test Case B
3	Macropore benchmark	Test Case C
4	Fractured domain	Test Case D
5	Mixed continua	Test Case E
6	Grid convergence	Numerical report
7	Performance analysis	Benchmark paper

WP1 – Climate & Surface Forcing (Expanded)

Objective

Develop a comprehensive climate and land-surface forcing framework capable of representing realistic spatial and temporal variability in atmospheric forcing, land surface characteristics, and contaminant loading for groundwater recharge simulations.

Stage	Task	Description	Expected Deliverable
WP1.1	Baseline Climate Framework	Define the minimum atmospheric forcing variables (P, ET, T, RH, wind, radiation)	Climate forcing framework
WP1.2	Homogeneous Rainfall	Uniform rainfall over the entire domain	Baseline rainfall module
WP1.3	Multi-Class Rainfall	Divide rainfall into rainfall zones or classes	Rainfall partition model
WP1.4	Spatial Rainfall Variability	Continuous spatially varying rainfall fields	Spatial rainfall generator
WP1.5	Temporal Rainfall Variability	Time-varying rainfall intensity throughout the simulation	Time-dependent rainfall module
WP1.6**	Event-Based Rainfall	Short-duration storm events	Storm event generator
WP1.7	Seasonal Climate Forcing	Wet season, dry season and transition periods	Seasonal forcing library
WP1.8	Stochastic Rainfall Generator	Random rainfall generation preserving statistical characteristics	Stochastic rainfall simulator

Stage	Task	Description	Expected Deliverable
WP1.9	Extreme Rainfall Events	Heavy rainfall, flash floods and rare storms	Extreme event module
WP1.10	Climate Change Scenarios	Future rainfall projections under different climate scenarios	Climate scenario database
WP1.11**	Evapotranspiration Module	Dynamic ET estimation	ET model
WP1.12	Boundary Condition Manager	Apply spatial and temporal forcing to WP2	Boundary condition interface
WP1.13	Sensitivity Analysis	Assess effects of rainfall characteristics on recharge	Sensitivity report
WP1.14	Validation	Compare forcing datasets with observations	Validation report

WP2

Develop a scalable, physically based multi-continuum vadose zone flow model that explicitly represents unresolved subsurface heterogeneity, preferential flow pathways, and continuum interactions to accurately predict infiltration, soil water redistribution, storage dynamics, and groundwater recharge under spatially and temporally varying environmental conditions.

Stage	Task	Deliverable
2.1	Single continuum Richards	Working solver
2.2	Dual continuum	Dual solver
2.3	Triple continuum	Triple solver
2.4	Generalized N continuum	Generic solver
2.5	Exchange coefficients	Parameter module
2.6	Capillary exchange	New physics
2.7	Cross diffusion	Multi-continuum solver
2.8	Rainfall activation	Dynamic activation
2.9	Preferential flow	Flow module
2.10	Storage dynamics	Storage module
2.11	Adaptive switching	Connectivity module

Stage	Task	Deliverable
2.12	Numerical optimization	Faster solver
2.13	Verification	Verification report
2.14	Benchmarking	Benchmark results
2.15	Publication	Journal paper

WP3 – Multi-Continuum Contaminant Transport

Objective

Develop a physically based contaminant transport model that accurately predicts contaminant migration through heterogeneous vadose zones and into groundwater while accounting for preferential flow, cross-continuum exchange, and multiple contaminant sources.

Stage	Task	Description	Expected Deliverable
WP3.1	Literature Review	Review contaminant transport in fractured and heterogeneous media	Literature review report
WP3.2	Conservative Solute Transport	Develop advection–dispersion transport without reactions	Base transport solver
WP3.3	Multi-Continuum Transport	Extend transport to N interacting continua	Multi-continuum transport model
WP3.4	Cross-Continuum Mass Exchange	Develop contaminant exchange between continua	Exchange module
WP3.5	Preferential Transport	Simulate contaminant bypass through macropores and fractures	Preferential flow transport module
WP3.6	Source-Term Framework	Define distributed, localized, transient and continuous contaminant sources	Source-term library
WP3.7	Surface Contamination Integration	Couple agricultural, landfill and urban contamination sources (WP1)	Surface contaminant interface
WP3.8	Human Activity Integration	Couple mining, pipelines, leakage and industrial contamination (WP5)	Anthropogenic source interface
WP3.9	Reactive Transport	Add sorption, degradation and decay processes	Reactive transport module

Stage	Task	Description	Expected Deliverable
WP3.10	Variable Source Scenarios	Implement seasonal and event-driven contaminant loading	Scenario generator
WP3.11	Numerical Stability Analysis	Evaluate convergence and numerical robustness	Stability report
WP3.12	Synthetic Benchmark Testing	Validate against analytical and published benchmark problems	Benchmark database
WP3.13	Sensitivity & Uncertainty Analysis	Identify dominant transport parameters	Sensitivity report
WP3.14	Model Verification	Compare against laboratory and published datasets	Verification report
WP3.15	Publication	Publish contaminant transport framework	Journal paper

WP4 – Aquifer System Dynamics

Objective

Develop an integrated groundwater model that links vadose-zone recharge, aquifer flow, groundwater storage, and contaminant migration.

Stage	Task	Description	Expected Deliverable
WP4.1	Aquifer Conceptualization	Define heterogeneous aquifer structures	Conceptual aquifer model
WP4.2	Recharge Coupling	Connect recharge outputs from WP2	Recharge interface
WP4.3	Groundwater Flow Solver	Implement Darcy flow model	Groundwater solver
WP4.4	Heterogeneous Hydraulic Properties	Incorporate spatially variable conductivity and storage	Property module
WP4.5	Boundary Condition Framework	Rivers, lakes, pumping wells and no-flow boundaries	Boundary-condition module
WP4.6	Aquifer Storage Dynamics	Simulate groundwater storage changes	Storage model
WP4.7	Contaminant Entry	Couple contaminant recharge from WP3	Contaminant recharge module

Stage	Task	Description	Expected Deliverable
WP4.8	Groundwater Transport	Simulate contaminant migration within aquifers	Aquifer transport solver
WP4.9	Groundwater Mixing	Simulate dilution and mixing	Mixing model
WP4.10	Travel-Time Analysis	Compute contaminant travel times	Travel-time module
WP4.11	Vulnerability Assessment	Identify vulnerable groundwater zones	Vulnerability maps
WP4.12	Regional Scaling	Extend from local to catchment scale	Regional simulation framework
WP4.13	Calibration	Estimate aquifer parameters using observations	Calibrated groundwater model
WP4.14	Validation	Compare predictions with field observations	Validation report
WP4.15	Publication	Publish integrated aquifer framework	Journal paper

WP5 – Anthropogenic & Engineered Perturbations

Objective

Represent human activities that modify groundwater quantity and quality through abstraction, infrastructure, land use, and engineered interventions.

Stage	Task	Description	Expected Deliverable
WP5.1	Human System Review	Review anthropogenic impacts on groundwater	Literature review
WP5.2	Groundwater Abstraction	Model pumping from wells	Pumping module
WP5.3	Water Demand Assessment	Domestic, agricultural and industrial demand	Demand model
WP5.4	Pipeline Leakage	Simulate potable water leakage	Leakage model
WP5.5	Sewer Leakage	Simulate wastewater leakage	Sewer contamination model
WP5.6	Industrial Discharges	Model industrial contaminant releases	Industrial source module

Stage	Task	Description	Expected Deliverable
WP5.7	Mining Activities	Simulate mine drainage and excavation impacts	Mining module
WP5.8	Oil & Gas Operations	Represent drilling and produced-water contamination	Oil & gas module
WP5.9	Engineered Recharge	Artificial recharge and infiltration systems	Managed recharge module
WP5.10	Irrigation Return Flow	Agricultural return flow processes	Irrigation module
WP5.11	Legacy Contamination	Historical contamination and delayed release	Legacy contamination module
WP5.12	Feedback Mechanisms	Couple groundwater response to human interventions	Feedback framework
WP5.13*	Future Development Scenarios	Population growth, urbanization and climate adaptation	Scenario library
WP5.14	Integrated Human Impact Assessment	Combine all anthropogenic perturbations	Integrated impact model
WP5.15	Publication	Publish human-groundwater interaction framework	Journal paper

WP6 – Potable Water Assessment & Decision Support

Objective

Develop an integrated decision-support system that assesses groundwater quantity, water quality, treatment needs, and long-term sustainability.

Stage	Task	Description	Expected Deliverable
WP6.1	Water Availability Assessment	Estimate groundwater availability	Water availability model
WP6.2	Water Quality Assessment	Compare groundwater quality with drinking-water standards	Water quality module
WP6.3	Potable Water Estimation	Combine quantity and quality constraints	Potable water estimator

Stage	Task	Description	Expected Deliverable
WP6.4	Treatment Assessment	Evaluate treatment options and effectiveness	Treatment assessment module
WP6.5	Water Safety Index	Develop integrated water safety indicator	Water safety index
WP6.6	Climate Change Assessment	Assess future climate impacts	Climate scenario module
WP6.7	Extreme Event Analysis	Evaluate droughts, floods and contamination events	Hazard assessment
WP6.8	Risk Mapping	Generate spatial groundwater risk maps	GIS risk products
WP6.9	Uncertainty Analysis	Quantify prediction confidence	Uncertainty report
WP6.10	Decision Support Engine	Develop rule-based decision framework	Decision-support engine
WP6.11	Visualization Dashboard	Interactive maps, plots and scenario tools	Dashboard prototype
WP6.12	Stakeholder Scenario Analysis	Compare management alternatives	Scenario analysis toolkit
WP6.13	Policy Recommendation Framework	Translate scientific outputs into policy guidance	Policy-support framework
WP6.14	Demonstration Case Studies	Apply the framework to selected catchments or regions	Demonstration report
WP6.15	Publication & Software Release	Publish results and release the integrated platform	Journal paper and software package